

Vulnerability Short Notes

What is Vulnerability?

A vulnerability is a weakness or flaw in a system, application, network, or process that attackers can exploit to gain unauthorized access or damage systems.

Examples: Weak password, Unpatched software, Open ports.

Basic Security Terms

Term	Definition
Vulnerability	Weakness in system
Threat	Potential danger/attacker
Exploit	Method used to attack vulnerability
Risk	Chance of damage caused by threat

Types of Vulnerabilities (Important Definitions)

1. Network Vulnerability

Weakness in network devices or configurations.

Examples: Open ports, Weak firewall, Unsecured protocols (Telnet, FTP)

2. Operating System Vulnerability

Weakness in OS due to missing patches or misconfiguration.

Examples: SMB flaws, Privilege escalation, Outdated Windows/Linux

3. Web Application Vulnerability

Weakness in web applications that attackers exploit through websites.

4. SQL Injection (SQLi)

Attacker inserts malicious SQL query into application input fields to access database.

Impact: Database theft, Login bypass

5. Cross-Site Scripting (XSS)

Injection of malicious JavaScript into web pages.

Type	Meaning
Stored XSS	Script stored in DB
Reflected XSS	Script reflected in response
DOM XSS	Executed through browser DOM

Impact: Cookie theft, Session hijacking

6. Cross-Site Request Forgery (CSRF)

Forces logged-in users to perform unwanted actions.

Example: Unauthorized money transfer.

7. File Inclusion Vulnerability

Type	Meaning
LFI	Local File Inclusion
RFI	Remote File Inclusion

Impact: Read sensitive files, Remote code execution

8. Command Injection

Attacker executes OS commands through vulnerable application.

Impact: Full server compromise

9. Broken Authentication

Weak authentication mechanisms allow unauthorized access.

Examples: Weak passwords, No MFA

10. Broken Access Control

Users access resources without permission.

Example: User accessing another user's account.

11. Security Misconfiguration

Improper security settings expose systems.

Examples: Default credentials, Debug mode ON

12. SSRF (Server-Side Request Forgery)

Attacker forces server to make internal requests.

Impact: Internal network access

13. Cryptographic Vulnerability

Weak encryption or hashing methods.

Examples: MD5, Weak SSL/TLS

14. Cloud Vulnerability

Weakness in cloud infrastructure.

Examples: Public S3 bucket, IAM misconfiguration

15. Wireless Vulnerability

Weakness in Wi-Fi security.

Examples: Weak Wi-Fi password, Rogue access point

16. Mobile Application Vulnerability

Weakness in Android/iOS apps.

Examples: Hardcoded credentials, Insecure APIs

17. IoT Vulnerability

Weakness in smart devices.

Examples: Default passwords, Open ports

18. Human Vulnerability

Security weakness caused by human mistakes.

Examples: Phishing, Social engineering

Vulnerability Severity

Severity	Meaning
Critical	Immediate threat
High	Serious risk
Medium	Moderate risk
Low	Minor issue

One-Line Quick Revision

- **Vulnerability:** Weakness
- **Threat:** Danger
- **Exploit:** Attack method
- **Risk:** Chance of damage
- **CVE:** Vulnerability ID
- **CVSS:** Severity score
- **SQLi:** Database attack
- **XSS:** Script injection
- **CSRF:** Forced user action
- **SSRF:** Server request attack
- **LFI/RFI:** File inclusion attack